

GEMINI AND NANO BANANA EXECUTION PLAYBOOK

Prompt Architecture, Editing Commands, and Quality Control for Gemini / Nano
Banana Image Generation and Editing

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For use with: AuthenticLens — The Realism Architect Gemini Gem
Confidential Reference Document

PURPOSE

Provide a Gemini-native execution system for image analysis, image generation, and image editing that is operationally aligned with current Gemini / Nano Banana behaviour.

NAMING NOTE

Use official model names in documentation and prompts. The user shorthand and official names are:

User Shorthand	Official Name	Notes
Nano Banana 2	Gemini 3.1 Flash Image Preview	Fast, efficient image generation and editing model
Nano Banana Pro	Gemini 3 Pro Image Preview	Higher-quality generation, more complex instructions
Nano Banana 2 Pro	Gemini 3.1 Flash Image family (general shorthand)	Clarify only if model selection is critical to the task

INTENDED GEMINI BEHAVIOUR

- Use direct, modular prompts — not giant keyword lists.
- Prefer targeted local edits over global regeneration.

- Preserve subject identity and composition unless explicitly told otherwise.
- Work conversationally in steps when the edit is complex.
- Keep multi-step edits in the same conversation whenever possible.
- Apply the quality-control checklist after every output.

THE 10-COMPONENT PROMPT ARCHITECTURE

Every prompt executed by the Gem must adhere to a strict 10-component formula, ensuring that the model processes intent, constraints, and physical variables systematically.

Component	Function	Example Implementation
1. Task Command	Defines the core operation to be executed.	"Enhance this photo for realism." / "Execute identity-preserving background replacement."
2. Preservation Constraints	Establishes absolute boundaries to prevent hallucination or identity drift.	"Do not alter facial landmarks, ethnicity, or apparent age. Keep body proportions exactly as provided."
3. Subject Description	Grounds the primary focus with physical, descriptive detail.	"A 35-year-old woman wearing a heavy wool trench coat, looking directly at the camera."
4. Scene/Background	Defines the environmental context and spatial depth.	"A busy urban intersection with historic limestone architecture and wet pavement."
5. Lighting	Dictates the key light direction, quality, and colour temperature.	"Overcast, highly diffuse skylight at 6000K, producing very soft, directionless shadows."
6. Camera/Lens Behaviour	Simulates specific optical physics and depth of field.	"Captured with a 50mm lens at f/2.8, creating shallow depth of field with circular bokeh."
7. Texture/Material Realism	Specifies micro-relief and specular behaviour for surfaces.	"Visible pores on the nose and cheeks; visible twill weave on the coat fabric."
8. Editing Style	Anchors the overall aesthetic to a specific photographic genre.	"Documentary photojournalism style, candid and unretouched."
9. Negative Constraints	Deploys targeted suppression terms to block algorithmic aesthetics.	"No AI glow, no plastic skin, no extra digits, no unreadable pseudo-text, no HDR halos."
10. Quality-Control Instruction	Instructs the model to self-verify before finalising the output.	"Verify exactly five fingers per hand and ensure all text is perfectly legible."

PRESERVATION-FIRST EDITING SYNTAX

Use these clauses whenever the request involves a real person:

- Preserve the subject's identity and recognisability.
 - Do not alter the face.
 - Keep the original skin complexion accurate.
 - Do not beautify aggressively.
 - Do not change clothing unless requested.
 - Do not change the background unless requested.
 - Only improve realism, lighting, texture, and photographic coherence.
 - Keep the same person, same pose logic, same age signal, and same facial structure.
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PROMPT MODULES BY USE CASE

Build prompts from these modular blocks. Combine as needed.

Realistic portrait enhancement

Enhance this portrait for realism only. Preserve the person exactly. Restore natural skin texture, believable pores, realistic eye detail, and soft hair edges. Reduce synthetic glow and over-smoothing. Keep lighting consistent and natural. No waxy skin, no identity drift.

Street-photo realism

Make this feel like a believable street photograph. Preserve the scene composition. Use realistic ambient light, grounded shadows, slight lens imperfections, authentic texture, and restrained colour. No fake bokeh, no impossible shadows.

Fashion editorial

Create a premium fashion-editorial finish while preserving the subject's identity. Keep garment drape believable, seams clean, and skin natural. Avoid plastic retouching and AI glow.

Luxury outfit swap

Change the outfit to a Saint Laurent-inspired luxury look while keeping the person fully recognisable. Preserve body proportions and pose. Use believable tailoring, realistic fabric weight, and minimal or perfectly readable branding. No melted fabric, no garbled logos.

E-commerce product photo

Improve this apparel product photo for premium e-commerce. Increase fabric and construction clarity, maintain true colour, clean the background, and keep the garment shape physically accurate. No oversharpening halos, no fake texture.

Background replacement

Replace the background only. Keep subject identity, pose, scale, and lighting logic. Match camera height, horizon, weather, season, and ground interaction to the new setting.

Group collage

Combine these subjects into one coherent group photo. Preserve every face. Match perspective, scale, grain, and light direction. Blend edges naturally and avoid identity drift.

AI-artifact correction

Correct AI-looking artefacts only. Repair hands, edges, text, fabric, and shadows without changing the subject's core appearance. No identity change, no restyling.

Lightroom-style realism

Give this a restrained Lightroom-style correction: balanced contrast, realistic white balance, controlled texture, subtle sharpening, and no overdone dehaze or HDR halos.

Phone-camera realism

Render this with believable modern phone-camera processing: restrained HDR, natural denoise, realistic sharpening, and plausible portrait-edge behaviour.

35mm film realism

Apply subtle 35mm-photo realism with natural grain, restrained halation only where justified, and authentic colour rolloff. Avoid exaggerated retro effects.

Ottawa spring realism

Place the scene in a believable Ottawa spring atmosphere: cool daylight (5500K-6500K), damp streets with Fresnel reflections, budding foliage (not full summer canopy), light jackets and trench coats, and realistic urban ground tones. Bilingual signage (STOP/ARRÊT) if street signs are visible.

Analyse authenticity

Analyse this image using four possible outcomes: (1) authentic real photo, (2) real photo with heavy editing, (3) AI-generated image, (4) composite/manipulated image. Separate structural AI artefacts from editing artefacts. Report: most likely class, authenticity confidence, heavy-edit confidence, AI confidence, composite confidence, uncertainty, and strongest evidence. Do not overclaim if the file quality is limiting.

NEGATIVE INSTRUCTION LIBRARY

Group negatives by failure type. Never use one giant undifferentiated block.

Failure Family	Negative Instructions
Identity drift	Do not change the face, identity, age signal, ethnicity, hairstyle, or expression.
Skin failures	No plastic skin, no waxy smoothing, no poreless cheeks, no extreme airbrushing, no doll-like face.
Anatomy failures	No warped hands, no distorted limbs, no fused fingers, no impossible joints, no extra digits.
Fabric failures	No melted fabric, no random wrinkles, no floating hems, no fake seam lines, no gravity-defying drape.
Light failures	No inconsistent shadows, no floating subject, no mismatched light direction, no fake rim light, no AI glow.
Background failures	No synthetic-looking background, no wrong scale, no mismatched perspective, no incorrect horizon.
Optical failures	No fake bokeh cutout, no excessive lens flare, no overused chromatic aberration.
Text/logo failures	No unreadable text, no garbled logos, no counterfeit-looking luxury branding, no pseudo-alphabetic text.
Processing failures	No over-smoothing, no over-sharpening halos, no overdone HDR, no crushed blacks unless requested, no render, no CGI, no digital art.

QUALITY-CONTROL CHECKLIST

After every output, verify:

- Did the face change?
 - Did the skin tone family shift?
 - Are the hands anatomically believable with exactly five digits?
 - Is the clothing physically plausible — fabric drape, wrinkle origin, seam continuity?
 - Do shadows connect the subject to the environment with correct direction?
 - Does the background match the subject's lens perspective and camera height?
 - Are logos readable, or should they have been simplified/omitted?
 - Does the image still look like one photograph, not several mixed images?
 - Has the realism pass become too obvious?
 - Is all visible text legible and semantically coherent?
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THE REPAIR LOOP

When the quality-control checklist reveals a failure:

1. Generate or edit the output.
2. Identify the visible failure clearly.

3. Classify the failure: identity, anatomy, light, texture, text, background, or processing.
 4. Rewrite the prompt with stricter local constraints targeting the specific failure.
 5. Apply targeted correction only — do not regenerate the entire canvas.
 6. Re-check the quality-control checklist.
 7. Stop before overprocessing.
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READY-TO-USE GEMINI PROMPT TEMPLATES

Enhance This Picture

Enhance this image for realism only.
Preserve the subject exactly as they are.
Do not change the face, skin tone family, age signal, hairstyle, body proportions, clothing, or background.
Improve only lighting balance, natural skin texture, realistic hair edges, shadow grounding, and photographic coherence.
Reduce any plastic smoothing or synthetic glow.
Keep the final result subtle and believable, like a real photograph.
No waxy skin, no identity drift, no impossible shadows.

Make This More Realistic and Less AI-Generated

Correct only the AI-looking artefacts in this image while preserving the original subject and composition.
Prioritise realistic hands, edges, shadows, skin texture, hairline detail, and fabric behaviour.
Reduce fake glow, over-smoothing, and synthetic bokeh.
Do not alter identity or restyle the image.
Keep the final result grounded in everyday photographic realism.

Change Background to Ottawa Spring

Keep the person fully recognisable and unchanged.
Replace the background only with a believable spring urban scene in Ottawa.
Match camera height, lens perspective, weather, lighting direction, and ground shadows.
Use cool daylight (5500K–6500K), damp pavement with Fresnel reflections, budding foliage, and realistic downtown spring atmosphere.
Do not change the clothing unless needed for light matching.
No synthetic-looking background, no mismatched perspective, no summer foliage.

Change Clothing to Saint Laurent While Keeping the Subject Natural

Keep the exact same person, face, hairstyle, pose, and body proportions.
Change the outfit to a realistic Saint Laurent-inspired luxury look with believable

tailoring,
premium fabric drape (grain de poudre wool or silk), and natural folds at articulation points.
Keep branding minimal or perfectly readable only if already present.
Do not make the result look like CGI fashion art.
Preserve realistic skin, lighting, and overall photographic credibility.
No melted fabric, no garbled logos, no identity drift.

Make This Product Photo Sharper for E-Commerce

Improve this apparel product image for premium e-commerce use.
Preserve the garment design and true colour.
Increase clarity of fabric weave, seams, stitching, zipper teeth, cuffs, hems, and silhouette.
Keep the lighting controlled, clean, and premium.
Avoid oversharpening halos, fake texture, or unnatural contrast.
Output should feel high-end, polished, and trustworthy for a web store.

Create a Realistic Group Collage Without Changing Faces

Combine these people into one believable group photo.
Preserve each face exactly.
Match perspective, scale, grain, lighting direction, and shadow density across all subjects.
Blend edges naturally and ground everyone into the same environment with contact shadows.
Do not beautify, restyle, or alter identity.

Analyse if This is Real, AI, Edited, or Composite

Analyse this image using four possible outcomes:

- 1) authentic real photo
- 2) real photo with heavy editing
- 3) AI-generated image
- 4) composite/manipulated image

Separate structural AI artefacts from editing artefacts.
Report: most likely class, authenticity confidence, heavy-edit confidence, AI confidence, composite confidence, uncertainty, and strongest evidence.
Do not overclaim if the file quality is limiting.
Return 'inconclusive' if uncertainty is above 45.

Fix Perspective and Alignment

Correct only perspective, horizon, vertical alignment, and grounding issues.
Do not change the subject, identity, lighting style, or composition intent.
Keep the result consistent with a real camera view.
Avoid over-straightening if a natural wide-angle look is appropriate.

Adjust Complexion from a Reference

Use the reference image only to guide complexion balance.
Keep the subject fully recognisable.
Make a subtle complexion adjustment only, consistent across face, neck, hands, and visible skin.
Do not change facial structure, age signal, or expression.
Keep the final colour believable under the scene's lighting conditions.
Correct white balance and exposure before adjusting skin tone.

QUICK-REFERENCE SUMMARY

- Use modular prompts, not giant keyword piles.
- Preserve first, edit second.
- Prefer local corrections over full re-renders.
- Use grouped negatives — categorise by failure type.
- Quality-check every output against the 10-point checklist.
- Iterate in small, targeted steps.
- Repair loop: identify failure > classify failure > targeted correction > re-check.
- 'Inconclusive' is valid when uncertainty is high.

INTEGRATION GUIDE — SIX-DOCUMENT STACK

The six documents work as one system:

Document 1	Legacy AI-artifact vocabulary and revised scoring	Primary reference for artifact identification
Document 2	Photographic and optical grounding	Physics reference for understanding what is real
Document 3	Historical realism-prompt vocabulary and Gemini prompting	Prompt construction reference
Document 4	Classification and diagnostic decision engine	Analyse with this — four-way classification
Document 5	Realism-enhancement operating manual	Plan edits with this — identity and physics hierarchy
Document 6	Gemini / Nano Banana execution layer	Phrase the final prompt or editing instruction through this

RECOMMENDED FILENAMES

- AuthenticLens_Doc1_AI_Artifact_Lexicon_Corrected.pdf
- AuthenticLens_Doc2_Photographic_Physics_Corrected.pdf
- AuthenticLens_Doc3_Realism_Prompting_Formulas_Corrected.pdf
- AuthenticLens_Doc4_Classification_Diagnostic_Framework.pdf
- AuthenticLens_Doc5_Realism_Enhancement_Protocols.pdf
- AuthenticLens_Doc6_Gemini_Nano_Banana_Execution_Playbook.pdf

END OF DOCUMENT 6 — GEMINI AND NANO BANANA EXECUTION PLAYBOOK
AuthenticLens Knowledge Base · For internal Gem use only